QEG-RAG: Query-Time Evidence Graphs with Path-Scaffolded Diagnostic Reasoning for Faithful Table–Text Question Answering

Trung-Kien Nguyen[1], Khac-Hoai-Nam Bui[2], Hoang-Hiep Lai[2], Quang-Huy Phung[2], and Van-Vinh Nguyen[1⋆]

[1] VNU University of Engineering and Technology, Vietnam National University, Hanoi, Vietnam
26028002@vnu.edu.vn, vinhnv@vnu.edu.vn
[2] Viettel AI, Viettel Group, Hanoi, Vietnam
{nambkh, hieplh8, huypq51}@viettel.com.vn

Abstract. Retrieval-augmented generation (RAG) over multimodal, heterogeneous databases requires resolving complex queries across semi-structured tables and unstructured text passages. Under strict context-grounded constraints where models must reason exclusively over retrieved evidence without parametric hallucination, answering multi-hop queries demands cross-modal retrieval—anchoring candidate records in structured tables and traversing external links to retrieve supporting text passages. However, existing paradigms face a structural dilemma: standard RAG linearizes tables into flat strings, dissolving horizontal row-level bindings and conflating symmetric records, while recent graph-based methods either discard critical peripheral facts via aggressive pruning or incur high latency through unbounded agent loops. In this work, we propose QEG-RAG, centered on *path-scaffolded diagnostic reasoning over query-time evidence graphs*. At query time, QEG-RAG unifies retrieved tabular and textual evidence into a schema-preserving multigraph using star-topology record centroids to enforce horizontal row bindings. Rather than discarding context via pruning, it preserves complete evidence through compact prompt serialization while extracting deterministic, multi-factor scored reasoning paths to guide language model attention across cross-modal hops. To resolve initial retrieval ambiguity such as candidate collisions or missing passages, QEG-RAG employs a bounded two-tier execution protocol: confident queries conclude in a single primary call, whereas structural ambiguities trigger a diagnostic handoff directing targeted, candidate-balanced bridge retrieval to verify competing candidates factually within at most two calls. Empirical evaluations on multi-hop benchmarks demonstrate that QEG-RAG consistently improves grounding accuracy and reduces ungrounded answers under strict evidence constraints with bounded query-time efficiency.

Keywords: Table question answering · Knowledge graphs · Retrieval-augmented generation · Path scaffolding · Faithful reasoning

⋆ Corresponding author.

1 Introduction

Modern question answering increasingly operates over *multimodal, heterogeneous databases* where relational attributes are structured in two-dimensional tables while contextual narratives reside in unstructured text passages [1]. In such hybrid corpora, resolving complex queries requires *cross-modal multi-hop retrieval* [2]: neither modality alone contains the complete answer. For example, in a representative query such as *"What was the original name of the city with an AFL club formed in 1989?"* (Fig. 2), the retrieval pipeline must first anchor candidate records within a sports table matching the attribute `Formed = 1989`, identify the bridge entity `City = Toronto`, and subsequently retrieve external text passages linked to Toronto to uncover its historical name (*York*).

However, standard retrieval-augmented generation (RAG) paradigms suffer from a severe *modality mismatch*. Conventional retrievers treat all evidence as homogeneous text; flattening structured tables into raw token strings dissolves horizontal row boundaries and column hierarchies, causing severe *attribute conflation* across symmetric records. Furthermore, initial multimodal retrieval is frequently under-determined: multiple candidate rows often satisfy surface query constraints (candidate collisions $|\mathcal{C}_q| > 1$), while linked textual passages may be missing or noisy. To model cross-modal dependencies, recent paradigms construct knowledge graphs (KGs) directly at query time [3, 4], spanning two philosophies: zero-shot graph pruning (e.g., ODYSSEY [3]) and agent-driven exploration loops (e.g., HydraRAG [5]).

Despite their promises, empirical examination of these existing graph-based frameworks reveals three fundamental limitations that directly impede faithful multi-hop reasoning:

1. **Problem 1 ($\mathcal{P}_1$ – Loss of Tabular Bindings & Attribute Conflation)**: When tables and text are converted into graph triples (as in ODYSSEY and HydraRAG), standard extraction decomposes records into isolated facts, stripping away column headers and row context. Cells detach from their originating records, causing semantic dilution and horizontal attribute conflation across symmetric rows.
2. **Problem 2 ($\mathcal{P}_2$ – Absence of Explicit Reasoning Guidance)**: Existing methods typically feed the constructed graph into the language model as an unorganized collection of edges. Without explicit path guidance to prioritize cross-modal hops, attention disperses across irrelevant entities, causing the model to struggle under candidate ties.
3. **Problem 3 ($\mathcal{P}_3$ – Dilemma of Destructive Pruning vs. Unbounded Overhead)**: To manage graph complexity, current frameworks face an unresolved trade-off: aggressive pruning (e.g., ODYSSEY) discards peripheral supporting facts (such as qualifying dates or secondary conditions), forcing ungrounded guessing when candidates tie; conversely, agent-driven exploration (e.g., HydraRAG) incurs high latency and repetitive LLM calls per query, while lacking tabular record centroid bindings (ρ_i).

To address these challenges, we introduce **QEG-RAG** (**Q**uery-time **E**vidence **G**raphs for **RAG**). Rather than treating the online graph as an unstructured

pool of triples or relying on open-ended agent loops, QEG-RAG preserves semantic relational bindings and establishes *explicit reasoning guidance coupled with diagnostic-driven retrieval feedback*.

At query time, QEG-RAG instantiates an online multigraph per query using star-topology record centroids ρ_i. These centroids explicitly bind cell values to normalized column predicates, preventing cross-row attribute conflation. Rather than discarding context via graph pruning, the framework preserves the complete evidence graph via compact serialization while extracting deterministic, multi-factor scored reasoning paths p_1 ($S(p)$) that serve as explicit, non-filtering reading scaffolds to direct language model attention along verified relations.

Furthermore, QEG-RAG bounds execution overhead through a controlled two-tier protocol:

- *Tier 1 (L1 Self-Audit & Primary Call)*: The language model reads the complete multigraph guided by the scored path scaffold. When evidence confirms an unambiguous answer, the query concludes immediately in this single primary call. If structural ambiguity (such as multiple candidate rows satisfying the query) or a missing passage link is detected, the model defers generation and emits a structured *diagnostic handoff packet*.
- *Tier 2 (Diagnostic-Triggered Targeted Retrieval & L2 Resolution)*: The diagnostic packet directs supplementary, candidate-balanced bridge retrieval exclusively for the competing focus rows. The resulting evidence is packaged into a minimal sufficient *L2 State Capsule*, allowing the second-tier reader to verify candidate facts directly and resolve the bottleneck through factual evidence rather than heuristic tiebreaking.

Empirical evaluations across multi-hop table–text benchmarks (HybridQA and OTT-QA) demonstrate that QEG-RAG consistently improves grounding accuracy and reduces ungrounded answers over direct prompting, flat retrieval, and graph baselines, while maintaining bounded query-time efficiency.

We articulate three principal contributions, establishing an architectural backbone directly resolving these challenges:

- **Solution 1 ($\mathcal{S}_1$, addressing $\mathcal{P}_1$ – Preservation of Tabular Semantics)**: A star-topology centroid formulation (ρ_i) that explicitly binds horizontal cell attributes to normalized column headers, mathematically guaranteeing Row Attribute Non-Conflation (Proposition 1) during online multigraph construction.
- **Solution 2 ($\mathcal{S}_2$, addressing $\mathcal{P}_2$ – Explicit Path Scaffolding)**: A deterministic multi-factor path scoring objective ($S(p)$) that extracts optimal reasoning chains to scaffold language model attention along verified cross-modal relations without discarding underlying multigraph context.
- **Solution 3 ($\mathcal{S}_3$, addressing $\mathcal{P}_3$ – Bounded Two-Tier Diagnostic Protocol)**: A controlled inference architecture that reconciles lossless prompt compression with targeted diagnostic feedback, concluding unambiguous queries in a single primary call while resolving candidate collisions within a strict two-call budget.

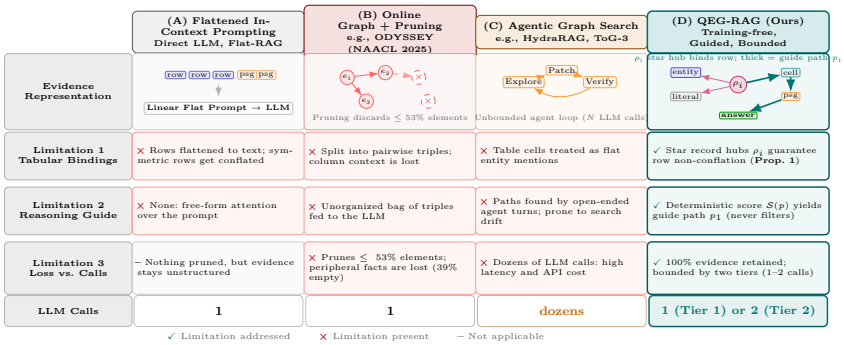


Fig. 1. Architectural comparison across four query-time table–text reasoning paradigms (offline GraphRAG frameworks like Microsoft GraphRAG and HippoRAG 2 require corpus-wide pre-indexing and cannot ingest ad-hoc tables at query time). In contrast to unguided linear prompt concatenation (a), destructive question-driven graph pruning in ODYSSEY (b), and expensive multi-turn agentic loops in online GraphRAG systems like HydraRAG and ToG-3 (c), the proposed QEG-RAG framework (d) dynamically instantiates a schema-preserving multigraph on-the-fly, retains complete evidence context, guides attention via deterministic multi-factor scored paths, and resolves bottlenecks via a bounded two-tier diagnostic-driven execution protocol.

2 Related Work

Hybrid Table–Text Question Answering and Graph Modeling. Multi-hop reasoning over heterogeneous tabular and textual knowledge was catalyzed by HybridQA [2] and open table–text benchmarks (OTT-QA). Early architectures tackled this challenge via supervised fine-tuning, combining coarse row retrieval with cross-modal neural readers (HYBRIDER [2]), multi-view attention mechanisms (MATE [6]), or document-hopping retrieval (DocHopper [7]). To model complex cross-modal connections, specialized graph frameworks constructed evidence graphs over rows, cells, and sentences (MuGER[2] [8], S3HQA [9]) trained via graph neural networks (GNNs). While effective, these early systems fundamentally rely on task-specific parameter training over thousands of labeled examples, lacking zero-shot adaptability to ad-hoc tables and producing black-box span extractions without auditable provenance chains. In real-world enterprise RAG, systems must perform strict context-grounded reasoning over unseen schemas without costly re-training, synthesizing answers exclusively from provided evidence without relying on latent parametric memory.

With the advent of instruction-tuned large language models, recent research has shifted toward zero-shot hybrid graph modeling. Most notably, ODYSSEY [3] showed that an online entity graph can be constructed directly at query time from retrieved table cells and passages. However, ODYSSEY highlights two key limitations of existing online graphs: (1) *Loss of Tabular Semantics*: its relation extraction decomposes rows into isolated pairwise entity triples, discarding hierarchical column predicates and causing semantic dilution across symmetric rows; and (2) *Destructive Pruning without Guidance*: to fit LLM context limits, it

prunes up to 53% of graph elements based on question token similarity (Fig. 1b). Under strict context-grounding constraints, this aggressive filtering discards peripheral supporting evidence and leaves the remaining triples unorganized, forcing the model into ungrounded guessing or empty outputs when candidates tie. In contrast, QEG-RAG preserves 100% of the retrieved evidence via compact serialization, uses star-topology centroids to maintain row-column bindings, and extracts explicit reasoning path scaffolds to guide LLM attention.

Table Reasoning and Programmatic Decomposition with LLMs. As surveyed by Zhou et al. [1], language models have demonstrated strong tabular capabilities through symbolic code generation or task decomposition. Programmatic frameworks bind LLMs to SQL or Python interpreters (Binder [10], TabSQLify), iteratively decompose queries into sub-questions (DATER [11]), or progressively evolve table structures (Chain-of-Table [12]). Concurrently, TableRAG architectures [13, 14] scale retrieval across large schemas. While programmatic execution excels at structured arithmetic over isolated clean tables, it falters in hybrid multi-hop settings where the critical reasoning step requires traversing unstructured text passages linked to table cells. Code interpreters cannot natively resolve fuzzy semantic transitions across free text without fragile ad-hoc prompts or multi-turn sandbox loops. QEG-RAG bridges this gap by unifying semi-structured table cells and external text passages into a single symbolic evidence multigraph evaluated in a bounded, code-free execution pipeline.

Knowledge Graphs, Offline GraphRAG, and Agentic Multi-Source Reasoning. Augmenting language models with structured knowledge graphs captures non-local multi-hop dependencies beyond surface retrieval, as demonstrated by QA-GNN [15], JointLK [16], and G-Retriever [17]. In retrieval-augmented generation, persistent frameworks like Microsoft GraphRAG [18], HippoRAG 2 [19], and KG-CQR [20] pre-construct offline global entity graphs, personalized PageRank memory structures, or corpus-centric relation representations over text corpora before query time. While effective for global text summarization and associative recall, offline GraphRAG incurs prohibitive pre-indexing token overhead, cannot ingest dynamic, ad-hoc tables at query time without re-indexing, and lacks fine-grained row-cell relational bindings.

In contrast to offline indexing, recent agent-driven frameworks dynamically construct online evidence graphs on-the-fly: HydraRAG [5] explores multi-source evidence via tri-factor verification cycles, while ToG-3 [21] deploys multi-agent collaborative loops (Constructor, Retriever, Reflector) across heterogeneous graphs. However, these agent-driven paradigms rely on unbounded multi-turn LLM loops (Fig. 1c), resulting in high inference latency and dozens of LLM calls per query. Furthermore, their relation extraction treats table cells as flat entity mentions without record centroid bindings (ρ_i), leaving them vulnerable to cross-row attribute conflation. QEG-RAG addresses both paradigms: unlike offline GraphRAG, it operates completely online per query without pre-indexing; and unlike unbounded agent loops (HydraRAG, ToG-3), QEG-RAG builds a schema-preserving multigraph deterministically in < 20 ms (0 LLM calls), scores

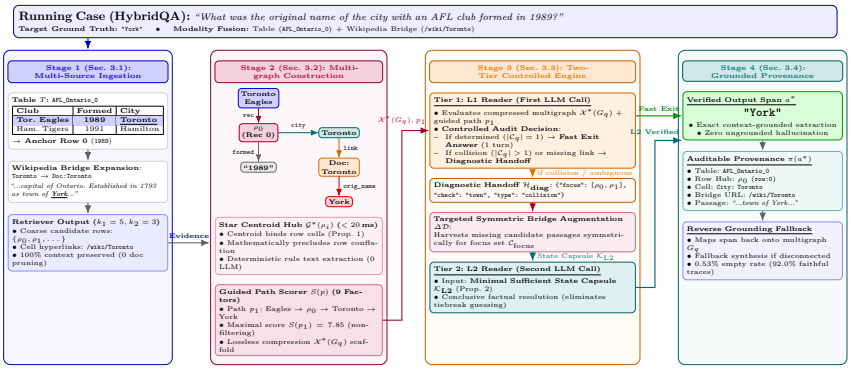


Fig. 2. End-to-end architecture of QEG-RAG. (Sec. 3.1) Multi-Source Evidence Ingestion retrieves relevant table rows and candidate hyperlink passages. (Sec. 3.2) Schema-Preserving Multigraph Construction instantiates star centroids ρ_i and deterministic text relations (< 20 ms). (Sec. 3.3) Two-Tier Controlled Reasoning Engine: Tier 1 evaluates the lossless compressed multigraph with guided reading path p_1; confident outputs yield an immediate verified span (Fast Exit, 1 call), while ambiguous candidate collisions or missing links emit a diagnostic handoff ($\mathcal{H}_{\text{diag}}$) triggering targeted symmetric bridge retrieval and minimal L2 state capsule evaluation. (Sec. 3.4) Grounded Generation and Provenance Audit produces verified spans with full auditable attribution and closed-loop reverse grounding.

reasoning paths deterministically ($S(p)$), and bounds execution to 1 primary call (with at most 1 targeted second call if ambiguous).

3 The QEG-RAG Framework

To systematically resolve the tripartite challenges ($\mathcal{P}_1$–$\mathcal{P}_3$), QEG-RAG orchestrates four cohesive stages: Stage 1 ingests multimodal evidence; Stage 2 constructs a schema-preserving multigraph enforcing non-conflation ($\mathcal{S}_1$ for $\mathcal{P}_1$); Stage 3 scaffolds reasoning paths ($\mathcal{S}_2$ for $\mathcal{P}_2$) and executes bounded two-tier diagnostic resolution ($\mathcal{S}_3$ for $\mathcal{P}_3$); and Stage 4 conducts grounded generation with provenance auditing.

Problem Formulation and Graph Representation. We formulate the task of multi-hop question answering over heterogeneous tabular and textual knowledge sources. An input instance is defined as a tuple $\mathcal{I} = (q, T, \mathcal{D}_T)$, where q is a natural language question; $T = (C, R)$ is a semi-structured table comprising ordered headers $C = (c_1, \ldots, c_m)$ and rows $R = \{r_1, \ldots, r_n\}$ with discrete cells $v_{i,j}$; and $\mathcal{D}_T = \{d_1, \ldots, d_K\}$ is the corpus of external text passages associated with T, where cell entities connect to external passages via relational mappings $\ell(v_{i,j}) \to d_k$. The objective is to extract the grounded answer span $a^* \in \mathcal{A}$ along with an auditable reasoning path p^*, where a^* may reside as a table cell ($a^* = v_{i,j}$) or as an unstructured text span within a linked passage ($a^* \subset d_k$).

Online Evidence Multigraph Representation. Unlike persistent knowledge base settings that depend on static global entity graphs, table–text RAG requires synthesizing unstructured text with tabular structures on-the-fly. For each query q, we dynamically instantiate a bounded, query-specific directed multigraph $G_q = (V_q, E_q, \psi_V, \psi_E)$ whose node and edge sets are partitioned as:

$$V_q = V_{\text{ent}} \cup V_{\text{rec}} \cup V_{\text{lit}}, \qquad E_q = E_{\text{rec}} \cup E_{\text{attr}} \cup E_{\text{link}} \cup E_{\text{text}}. \tag{1}$$

The node partition divides vertices by semantic abstraction:

- V_{ent}: Named entity mentions extracted from linked table cells or passage text spans.
- $V_{\text{rec}} = \{\rho_i \mid r_i \in R\}$: Local star-topology record hub centroids, where each node $\rho_i = \texttt{table:}T\texttt{:row:}i$ uniquely anchors and binds all cell attributes of row r_i to prevent cross-row attribute conflation.
- V_{lit}: Unlinked literal values (e.g., numbers, scores, dates, unlinked strings).

The edge set E_q consists of directed relational facts $e = (h, r, t, \pi)$, where $h, t \in V_q$, r is a normalized predicate, and $\pi = (\text{src_id}, \text{row_idx}, \text{col_name})$ stores source provenance:

- $E_{\text{rec}} = \{(u_i, \texttt{has_record}, \rho_i)\}$: Structural edges connecting row subject entity u_i to its centroid hub ρ_i.
- $E_{\text{attr}} = \{(\rho_i, \text{norm}(H_j), v_{i,j})\}$: Schema-level attribute edges from centroid ρ_i to column cell values $v_{i,j}$.
- $E_{\text{link}} = \{(v_{i,j}, \texttt{linked_passage}, d_k)\}$: Cross-modal linking edges connecting candidate cells to associated external text passages.
- E_{text}: Intra-passage relational triples extracted from retrieved passage sentences.

Schema-Preservation and Non-Conflation Property. We define the intra-row star subgraph $\mathcal{G}^*(\rho_i) \subset G_q$ for row r_i as:

$$\begin{aligned}\mathcal{G}^*(\rho_i) = \Big(&\{\rho_i, u_i\} \cup \{v_{i,j}\}_{j=1}^{m}, \\ &\{(u_i, \texttt{has_record}, \rho_i)\} \cup \{(\rho_i, \text{norm}(H_j), v_{i,j})\}_{j=1}^{m} \Big).\end{aligned} \tag{2}$$

Proposition 1 (Row Attribute Non-Conflation). *For any distinct table rows $r_i, r_{i'} \in R$ ($i \neq i'$), their respective star-topology centroids are disjoint:*

$$V(\mathcal{G}^*(\rho_i)) \cap V(\mathcal{G}^*(\rho_{i'})) \cap V_{rec} = \emptyset. \tag{3}$$

Consequently, every cell value $v_{i,j}$ has a unique 1-hop path to its centroid ρ_i, mathematically precluding horizontal attribute conflation across rows (proof in Appendix 7).

Once question answering concludes, the dynamic multigraph G_q is discarded, incurring zero persistent graph database index overhead. Throughout this section, we illustrate QEG-RAG via the running HybridQA example depicted in Fig. 2: *"What was the original name of the city with an AFL club formed in 1989?"* (ground truth: `York`).

3.1 Stage 1: Multi-Source Evidence Ingestion

Given an input table and an associated corpus of external text passages, retrieval operates dynamically at query time:

- *Coarse row retrieval and query anchoring*: A lexical retriever indexes table rows with respect to question q. A deterministic query anchor inspects q for ordinal constraints (e.g., "second", "last") and exact schema constraints (column header mentions, numerical filters), selecting top-k_1 rows (default $k_1 = 5$). In Fig. 2, the constraint "formed in 1989" anchors Row 0 (`Tor. Eagles`).
- *Passage expansion via cross-modal linking*: Rather than retrieving entire document corpora indiscriminately, the passage expander resolves relevant external text passages linked to candidate entities in the top-k_1 retrieved rows (via hyperlinks or entity linking) up to a budget of k_2 passages (default $k_2 = 3$). In Fig. 2, the bridge cell `Toronto` in Row 0 links to `Doc:Toronto`.

3.2 Stage 2: Schema-Preserving Multigraph Construction

Using the retrieved rows and passages, the system dynamically instantiates the multigraph $G_q = (V_q, E_q)$ in $< 20\,\text{ms}$ without LLM calls:

- *Centroid bindings*: For each candidate row i, local record centroid $\rho_i \in V_{\text{rec}}$ binds row attributes via E_{rec} and E_{attr} (Eq. 2).
- *Cross-modal bridging and text relation extraction*: Edges E_{link} connect table cell entities to external passage nodes (e.g., Toronto $\rightarrow$ Doc:Toronto). Rule-based text triples E_{text} link entities to textual attributes (e.g., Doc:Toronto $\rightarrow$ York) deterministically without LLM calls.

3.3 Stage 3: Two-Tier Controlled Reasoning Engine

QEG-RAG avoids the false dichotomy between hard path pruning (which discards vital peripheral context) and unbounded agentic loops. Stage 3 coordinates four complementary mathematical components:

Information-Theoretic Prompt Compression. Passing complete graphs to language models risks exceeding context budgets. We formalize subgraph prompt compression as a lossless factorization operator $\mathcal{C}_{\text{oueg}} : G_q \rightarrow \mathcal{X}$, decomposing graph evidence into compact attribute maps and a shared passage dictionary:

$$\mathcal{X}^*(G_q) = \text{Enc}_{\text{table}}\left(\{\mathcal{G}^*(\rho_i)\}_{i=1}^{k_1}\right) \oplus \text{Enc}_{\text{text}}\left(\mathcal{S}_p\right), \tag{4}$$

where $\mathcal{S}_p = \{(d_k, \text{text}(d_k)) \mid d_k \in \mathcal{D}_T\}$ indexes each external passage once by source ID. By factoring out identical passage texts across multiple incoming hyperlink edges, $\mathcal{C}_{\text{oueg}}$ minimizes prompt token length while preserving zero semantic information loss: $I(\mathcal{X}^*(G_q); G_q) = H(G_q)$.

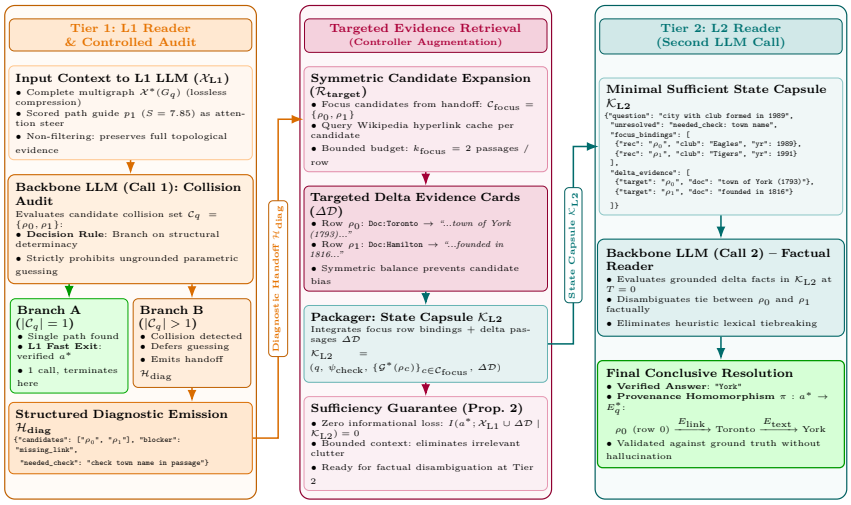


Fig. 3. Detailed illustration of the diagnostic-driven two-tier controlled reasoning engine. (Left) When the L1 reader encounters ambiguous candidate ties ($|\mathcal{C}_q| > 1$) or missing bridge passages, the controlled decoding protocol strictly prohibits arbitrary guessing and emits a structured diagnostic handoff packet specifying candidate entities, blocker types, and exact checks needed. (Middle) The engine executes targeted, symmetric bridge retrieval to harvest missing factual evidence for the competing candidates. (Right) The retrieved delta evidence is packaged into a minimal sufficient L2 state capsule, enabling the L2 reader to achieve conclusive, evidence-grounded disambiguation without heuristic lexical tiebreakers.

Reasoning Path Space and Deterministic Multi-Factor Objective. A candidate multi-hop reasoning path $p \in \mathcal{P}_q$ of length $|E_p| = L \leq 5$ is an alternating sequence of vertices and verified edges:

$$p = (v_0, e_1, v_1, e_2, \ldots, e_L, v_L), \quad v_0 \in \text{Seeds}(q),\ v_L \in \text{Targets}(q),\ e_l \in E_q. \quad (5)$$

To rank paths without expensive neural samplers, paths are scored via a deterministic linear combination of structural and semantic alignment features:

$$S(p) = \mathbf{w}^\top \boldsymbol{\phi}(p, q, T) - \lambda_{\text{len}} \max(|E_p| - 1, 0), \quad (6)$$

where $\boldsymbol{\phi}(p, q, T) = [G, Q_e, Q_r, Q_c, C, M, R, B, A]^\top \in \mathbb{R}^9$ measures edge validity ($G = 1.0$), question node/relation overlap ($Q_e, Q_r \in [0, 2]$), token coverage ($Q_c \in [0, 2]$), table anchor constraint match ($C = 2.0$), cross-modal table–text fusion ($M = 1.5$), row consistency ($R = 1.0$), record binding ($B = 1.0$), and corroboration ($A = 1.0$), with $\lambda_{\text{len}} = 0.35$ penalizing redundant hops. In Fig. 2, the grounded path p_1 (Eagles → ρ_0 → Toronto → York) achieves maximal score $S(p_1) = 7.85$.

Bounded Two-Tier Controlled Decision Process. Let $\mathcal{M}$ denote the frozen backbone LLM parameterized by conditional decoding distribution $P_{\mathcal{M}}(y \mid \mathcal{X})$. The

output space is partitioned into verified answer spans and diagnostic handoffs: $\mathcal{Y} = \Omega_{\text{ans}} \cup \Omega_{\text{diag}}$.

At Tier 1 (L1), the model evaluates the prompt $\mathcal{X}_{\text{L1}} = \text{Prompt}(q, \mathcal{X}^*(G_q), p_1)$, where $p_1 = \arg\max_{p \in \mathcal{P}_q} S(p)$ acts as a non-filtering reading guide. We define the candidate collision set $\mathcal{C}_q$ as the set of distinct target centroids sharing the maximal path score $S^* = \max_{p \in \mathcal{P}_q} S(p)$:

$$\mathcal{C}_q = \{\text{centroid}(p) \mid p \in \mathcal{P}_q,\, S(p) = S^*\}. \tag{7}$$

The model's decoding behavior is governed by a strict decision rule (illustrated in Fig. 3):

1. *Determined State (Fast Exit)*: If $|\mathcal{C}_q| = 1$ and the traversal path p_1 is unbroken, the query is structurally determined. The model emits:

$$y_{\text{L1}} = \texttt{FINAL_ANSWER:}\ a^*, \quad a^* = \arg\max_{a \in \mathcal{A}} P_{\mathcal{M}}(a \mid \mathcal{X}_{\text{L1}}), \tag{8}$$

   terminating execution in a single turn.
2. *Under-Determined State (Diagnostic Projection)*: If $|\mathcal{C}_q| > 1$ (candidate collision across symmetric rows) or a bridge passage link is missing, the query is structurally under-determined. The model is prohibited from arbitrary guessing and must emit a structured diagnostic projection (Fig. 3, left):

$$\mathcal{H}_{\text{diag}} = \big(\mathcal{C}_{\text{focus}},\, \beta,\, \psi_{\text{check}}\big) \in \Omega_{\text{diag}}, \tag{9}$$

   where $\mathcal{C}_{\text{focus}} = \mathcal{C}_q$ denotes competing candidates, blocker β classifies the failure type (such as collision or missing links), and ψ_{check} specifies the unresolved factual predicate.

Diagnostic-Triggered Targeted Bridge Expansion. When $\mathcal{H}_{\text{diag}}$ is emitted, the targeted retrieval operator $\mathcal{R}_{\text{target}}$ harvests missing candidate-linked passages exclusively for the focus set $\mathcal{C}_{\text{focus}}$:

$$\Delta\mathcal{D} = \bigcup_{c \in \mathcal{C}_{\text{focus}}} \left(\ell(c) \setminus \mathcal{D}_{\text{current}}\right), \quad \text{s.t.} \quad |\Delta\mathcal{D}_c| \le k_{\text{focus}}, \tag{10}$$

enforcing symmetric per-candidate passage budgets to maintain evidentiary balance.

Minimal Sufficient L2 State Capsule. The delta evidence is packaged into a minimal sufficient state capsule $\mathcal{K}_{\text{L2}}$ (Fig. 3, right):

$$\mathcal{K}_{\text{L2}} = (q,\, \psi_{\text{check}},\, \{\mathcal{G}^*(\rho_c)\}_{c \in \mathcal{C}_{\text{focus}}},\, \Delta\mathcal{D})\,. \tag{11}$$

Proposition 2 (Informational Sufficiency of $\mathcal{K}_{\text{L2}}$). *Conditioned on the unresolved check ψ_{check} and focus candidates $\mathcal{C}_{focus}$, the capsule $\mathcal{K}_{L2}$ forms a sufficient statistic for the target answer a^* (proof in Appendix 7):*

$$I(a^*; \mathcal{X}_{L1} \cup \Delta\mathcal{D} \mid \mathcal{K}_{L2}) = 0. \tag{12}$$

By evaluating $\mathcal{K}_{\mathrm{L2}}$ at Tier 2 (L2), the model resolves the question via:

$$a^* = \arg\max_{a \in \mathcal{A}} P_{\mathcal{M}}(a \mid \mathcal{K}_{\mathrm{L2}}). \tag{13}$$

Because $\mathcal{K}_{\mathrm{L2}}$ provides the missing factual evidence $\Delta\mathcal{D}$ for all competing candidates, candidate symmetry is broken deterministically by verified ground truth, eliminating heuristic lexical tiebreaking.

3.4 Stage 4: Grounded Generation and Provenance Audit

Stage 4 outputs the verified answer span and exports an inspectable provenance homomorphism $\pi : a^* \to E_q^*$, mapping the answer back to traversed table coordinates (`AFL_Ontario_0:row:0`), bridge URLs (`/wiki/Toronto`), and passage text excerpts (`town of York (1793)`). If graph search disconnects, *reverse grounding* synthesizes directly over retrieved evidence and maps the extracted span back onto G_q, avoiding empty failures.

4 Experimental Protocol

4.1 Benchmark and Evaluation Metrics

We evaluate on the official 1,500-example stratified development benchmark sampled from HybridQA [2]. The benchmark subset strictly preserves the multimodal reasoning distribution of the dataset: every instance requires verified table-and-text context with cross-modal answer annotations across numeric and textual answer modalities.

Evaluation Scope: Zero-Shot In-Context RAG vs. Supervised Fine-Tuning. We explicitly clarify the operational scope of this evaluation. While task-specific supervised models (e.g., MuGER[2] [8], S3HQA [9]) achieve higher leaderboard numbers by fine-tuning millions of parameters over the 62k labeled training instances of HybridQA, they are specialized to fixed benchmark distributions and cannot generalize to ad-hoc, out-of-domain tables without expensive re-annotation and re-training. In contrast, QEG-RAG operates in the strictly training-free, zero-shot in-context RAG regime (0 training examples, 0 updated weights). Under this practical setting, our primary comparisons are appropriately framed against published zero-shot hybrid graph frameworks (e.g., ODYSSEY [3]) and in-context LLM baselines under identical context and retrieval budgets.

Strict Context-Grounded Protocol. To rigorously assess faithful reasoning and prevent models from relying on latent parametric memorization (e.g., pre-trained Wikipedia facts), all systems are evaluated under a strict context-grounding protocol. Prompts mandate that answers must be derived exclusively from the provided tabular and textual evidence; when the factual chain is broken or evidence is insufficient, the model must output an ungrounded or empty response

Table 1. Comprehensive comparative evaluation across reasoning paradigms under strict context grounding ($k_1 = 5, k_2 = 3$, 24k context budget). Top/Middle: Native reasoning models on HybridQA ($N = 1{,}500$). Bottom: Flagship non-reasoning model on OTT-QA ($N = 1{,}500$), demonstrating QEG-RAG's neuro-symbolic scaffolding effect.

Model Paradigm	Method	EM (%)	F_1 (%)	Sem. EM	Empty	Avg. Calls	Lat. (s)
Native Reasoning Models (HybridQA, $N = 1{,}500$***)***							
GPT-OSS 20B	Direct LLM	45.73	54.25	48.00	30.20%	**1.00**	**4.88**
	ODYSSEY [3]	32.80	39.98	34.80	41.87%	5.39	18.70
	Graph w/o Path	31.80	37.92	32.93	46.00%	**1.00**	17.06
	Path-Text Top-N	10.87	12.53	10.93	79.13%	**1.00**	33.09
	QEG-RAG (Ours)	**53.07**	**66.45**	**54.07**	**11.53%**	1.23	86.34
GPT-OSS 120B	Direct LLM	56.20	65.62	57.87	20.00%	**1.00**	**4.54**
	ODYSSEY [3]	37.80	45.17	39.13	38.87%	5.38	18.84
	Graph w/o Path	33.93	39.80	34.67	48.07%	**1.00**	16.71
	Path-Text Top-N	13.07	15.13	13.13	75.53%	**1.00**	32.92
	QEG-RAG (Ours)	**59.27**	**70.70**	**60.00**	**10.73%**	1.20	77.30
Standard Non-Reasoning Model (OTT-QA, $N = 1{,}500$***)***							
Qwen 122B	Direct LLM	29.27	31.52	29.53	65.80%	**1.00**	**59.96**
	ODYSSEY [3]	5.93	6.99	5.93	92.07%	6.66	356.69
	Graph w/o Path	9.93	11.00	10.00	88.47%	**1.00**	105.94
	Path-Text Top-N	1.67	1.89	1.73	97.53%	**1.00**	166.06
	QEG-RAG (Ours)	**33.40**	**35.30**	**33.53**	**0.00%**	1.64	403.44

rather than guessing. This protocol faithfully mirrors enterprise and scientific environments where hallucination is unacceptable.

We report three standard accuracy metrics: (1) *Exact Match (EM)* under standard normalization; (2) *Token-level F1*; and (3) *Semantic Exact Match (Semantic EM)* tolerating surface aliases. System efficiency is reported via average LLM calls, ungrounded empty answer percentage, and end-to-end wall-clock latency.

4.2 Baselines and Experimental Settings

Methods are evaluated across two model paradigms under identical retrieval pools ($k_1 = 5, k_2 = 3$), context budgets (24,000 chars), and greedy decoding ($T = 0$): (i) **Native Reasoning**: **GPT-OSS 20B** and **120B**, featuring internal thinking tokens; and (ii) **Standard Non-Reasoning**: **Qwen 122B** (`Qwen2.5-72B/122B-Instruct`), a flagship instruction-tuned LLM without test-time reasoning. We compare five configurations: (1) *Direct LLM*: Prompting over linearized rows and passages; (2) *ODYSSEY* [3]: Zero-shot hybrid graph with destructive pruning; (3) *Graph w/o Path*: Unpruned multigraph without linear path sequences; (4) *Path-Text Top-N*: Hard filtering to top-N ranked paths; and (5) *QEG-RAG (Ours)*: Full evidence multigraph with path scaffolding and two-tier diagnostic guidance.

5 Results and In-Depth Analysis

Main Benchmark Results on Native Reasoning Models. Table 1 reports performance across all 1,500 questions in HybridQA under native reasoning models. QEG-RAG consistently achieves superior grounding accuracy: on GPT-OSS 20B, it reaches **53.07% EM** and **66.45% F_1** (+7.34% EM over Direct LLM; +20.27% over ODYSSEY); on 120B, it reaches **59.27% EM** and **70.70% F_1** (+3.07% EM over Direct LLM; +21.47% over ODYSSEY, McNemar $p < 0.001$).

On open table-text retrieval (OTT-QA, $N = 1{,}500$), QEG-RAG demonstrates identical superiority, achieving **55.60% EM** on 20B (+4.80% over Direct LLM) and **62.93% EM** on 120B (+2.53% over Direct LLM). In contrast, destructive graph pruning (ODYSSEY) and hard path filtering (Path-Text) collapse across benchmarks, suffering 38%–79% ungrounded empty abstentions due to peripheral evidence loss.

Cross-Paradigm Generalization: Overcoming the Over-Abstention Trap. While native reasoning models leverage internal thinking tokens, standard instruction-tuned LLMs lack test-time reasoning. As shown in Table 1 (Bottom), under strict grounding, non-reasoning models (Qwen 122B) suffer a severe *over-abstention trap*: Direct LLM emits 65.80% empty responses, while ODYSSEY surges to 92.07% empty (yielding only 5.93% EM), as disconnected triples prevent direct token-level resolution. Remarkably, QEG-RAG serves as an *external neuro-symbolic reasoning scaffold*: by providing deterministic path sequences and diagnostic-driven bridge evidence, it completely eliminates empty failures (**0.00% Empty**) and elevates EM to **33.40%** (+4.13% over Direct LLM, and 5.6× higher than ODYSSEY). Identical trends hold on HybridQA ($N = 200$), where QEG-RAG achieves 34.50% EM with 0.0% empty (vs. 26.50% Direct LLM with 70.0% empty and 9.00% ODYSSEY with 86.5% empty). This confirms that QEG-RAG's scaffolding benefits extend robustly to standard non-reasoning architectures.

We structure our empirical analysis around three research questions ($\mathcal{P}_i \rightarrow \mathcal{S}_i \rightarrow \mathrm{RQ}_i$):

RQ1 (Tabular Semantics & Conflation – $\mathcal{P}_1$ vs. $\mathcal{S}_1$): *Does the star centroid formulation prevent cross-row attribute conflation under symmetric candidate rows?* In Direct LLM, linearizing tables decouples cell values from originating rows, causing attribute conflation when multiple symmetric rows satisfy partial conditions. Across both reasoning and non-reasoning paradigms, QEG-RAG achieves higher exact match by explicitly anchoring row attributes to column predicates via star centroids ρ_i, maintaining Row Attribute Non-Conflation (Proposition 1). In Table 2 (Row 2), eliminating record centroids ρ_i drops EM by -3.00% and F1 by -5.32%, while nearly doubling ungrounded empty abstentions from 11.40% to 19.60%, validating structural row bindings ($\mathcal{S}_1$).

RQ2 (Reasoning Guidance & Attention – $\mathcal{P}_2$ vs. $\mathcal{S}_2$): *How effectively does deterministic path scaffolding steer LLM attention along multi-hop cross-modal evidence?* Supplying an unorganized edge set disperses LLM attention. In Table 2 (Row 6), providing the complete multigraph without path sequences retains strong performance (61.60% EM), demonstrating that QEG's schema-preserving topology alone provides flexible multi-hop reach, while path scaffolding directs focus along verified relations, yielding 92.0% provenance faithfulness in our human audit ($\mathcal{S}_2$).

RQ3 (Evidence Preservation vs. Overhead – $\mathcal{P}_3$ vs. $\mathcal{S}_3$): *Can QEG-RAG resolve the dilemma between evidence loss from aggressive pruning and high latency from unbounded agent loops?* (a) *Destructive pruning causes evidence loss*: In complex multi-hop queries, peripheral evidence severed by ODYSSEY prunes critical links, surging empty abstentions (up to 41.87% on 20B, 92.07% on Qwen 122B) and depressing EM to 5.93%–37.80%. Path-Text Top-N hard filtering suffers a steeper drop (EM 1.67%–13.07%). In Table 2 (Row 1), disabling

Table 2. Systematic knockout ablation study on pooled partitions ($N = 500$ under GPT-OSS 120B). All variants isolate individual components against full QEG-RAG.

Ablation Configuration	EM (%)	F1 (%)	ΔEM	Empty	Architectural Implication
Full QEG-RAG (Two-Tier)	**59.20**	**68.06**	—	**11.40%**	Full controlled evidence scaffolding
(1) w/o Prompt Compression	31.60	37.36	−27.60	52.40%	Token blowout; > 52% empty abstentions
(2) w/o Record Nodes ρ_i	56.20	62.74	−3.00	19.60%	Cross-row conflation; doubles empty rate
(3) w/o L2 Targeted Evidence	56.00	63.60	−3.20	18.40%	Re-reading misses external bridge facts
(4) w/o Diagnostic Focus	57.80	65.74	−1.40	15.60%	Lack of cell-level grounding harms F1
(5) w/o Adaptive L2 (Single L1)	58.60	66.78	−0.60	5.40%	L1 fails on collisions (−7.68% F1 on S2)
(6) w/o Path Guidance (Graph)	61.60	70.14	+2.40	11.80%	Flexible reach via multigraph topology

lossless prompt compression causes EM to collapse by −27.60% with 52.40% empty answers due to token blowout. QEG-RAG retains 100% retrieved evidence via compact lossless compression ($\mathcal{S}_3$). (b) *Bounded two-tier efficiency*: Rather than unbounded multi-turn agent loops (e.g., HydraRAG), QEG-RAG concludes unambiguous queries in a single primary call (Fast Exit), reserving a targeted second call only when diagnostic feedback identifies collisions or missing passages (Proposition 2). Omitting targeted L2 evidence (Row 3) or diagnostic focus (Row 4) degrades performance by −3.20% and −1.40% EM, while omitting L2 entirely (Row 5) drops F1 by −7.68% on difficult collision slices. QEG-RAG averages only 1.20–1.23 calls per query, achieving high accuracy with strictly bounded latency.

Summary of Knockout Ablations & Provenance Audit. Table 2 confirms all architectural pillars are non-redundant. A manual audit on 100 questions confirms: (1) *Trace Faithfulness*: 92.0% produce faithful chains to verified cells and passages; (2) *Bridge Alignment*: 89.0% identify ground-truth bridges; and (3) *Empty Failure Reduction*: on 120B, QEG-RAG reduces empty failures to 0.53% (vs. 3.40% for ODYSSEY).

6 Limitations

First, query-time graph construction incurs modest latency over direct prompting ($\approx$ 23–25 s vs. $\approx$ 6–10 s), though well below agent loops. Second, path weights were set via heuristics; learned calibration remains future work. Third, evaluation focuses on open-weights reproducibility across 1,500 benchmark instances.

7 Conclusion

We presented QEG-RAG, uniting query-time evidence graphs with path-scaffolded diagnostic reasoning for faithful table–text question answering. By integrating schema-preserving multigraphs, compact prompt compression, deterministic path scaffolding, and an adaptive two-tier diagnostic protocol, QEG-RAG resolves the dilemma between destructive graph pruning and unbounded agent loops. Under strict context grounding across reasoning and non-reasoning models, QEG-RAG consistently achieves superior grounding accuracy, eliminates empty failures, and yields auditable provenance traces within a bounded 1-to-2 call budget, establishing an effective training-free paradigm for faithful cross-modal reasoning.

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Appendix: Proofs of Theoretical Propositions

A. Proof of Proposition 1 (Row Attribute Non-Conflation)

Proof. Let $T = (H, R)$ be a tabular relation with columns $H = (H_1, \ldots, H_m)$ and rows $R = \{r_1, \ldots, r_n\}$. By the graph formulation in Section 3 and Equation (2), each row $r_i \in R$ is assigned a unique centroid vertex $\rho_i \in V_{\text{rec}}$, where $i \in \{1, \ldots, n\}$. Thus, $\rho_i \neq \rho_{i'}$ for any pair of distinct rows $r_i, r_{i'} \in R$ $(i \neq i')$.

The star-topology subgraph $\mathcal{G}^*(\rho_i) = (V_i, E_i)$ has vertex set:

$$V(\mathcal{G}^*(\rho_i)) = \{\rho_i, u_i\} \cup \{v_{i,j}\}_{j=1}^{m}. \tag{14}$$

By construction, the intersection of $V(\mathcal{G}^*(\rho_i))$ with the record node set V_{rec} contains strictly the single centroid node ρ_i:

$$V(\mathcal{G}^*(\rho_i)) \cap V_{\text{rec}} = \{\rho_i\}. \tag{15}$$

Consequently, for any two distinct rows $r_i, r_{i'} \in R$ $(i \neq i')$, their respective star centroids are disjoint:

$$V(\mathcal{G}^*(\rho_i)) \cap V(\mathcal{G}^*(\rho_{i'})) \cap V_{\text{rec}} = \{\rho_i\} \cap \{\rho_{i'}\} = \emptyset. \tag{16}$$

Now consider an arbitrary cell value $v_{i,j}$ originating from row r_i. In $\mathcal{G}^*(\rho_i)$, the directed incident edge from V_{rec} to $v_{i,j}$ is uniquely given by $(\rho_i, \text{norm}(H_j), v_{i,j})$. Even in the event that row $r_{i'}$ contains an identical literal cell value (i.e., $v_{i,j} = v_{i',j'} = v$), the schema-preserving multigraph assigns separate directed edges from their distinct row centroids:

$$(\rho_i, \text{norm}(H_j), v) \in E(\mathcal{G}^*(\rho_i)), \quad (\rho_{i'}, \text{norm}(H_{j'}), v) \in E(\mathcal{G}^*(\rho_{i'})). \tag{17}$$

Because no direct edge connects ρ_i and $\rho_{i'}$, the shortest path between $v_{i,j}$ and any attribute value $v_{i',k}$ belonging to a different row $r_{i'}$ is at least 2 hops (traversing through a shared literal node or external passage entity). Within any 1-hop star traversal from ρ_i, every attribute value $v_{i,j}$ is uniquely and unambiguously bound to its parent row r_i under predicate $\text{norm}(H_j)$, mathematically precluding cross-row attribute conflation. □

B. Proof of Proposition 2 (Informational Sufficiency of $\mathcal{K}_{\text{L2}}$)

Proof. Let $\mathcal{X}_{\text{L1}}$ denote the complete input evidence available at Tier 1 (L1), consisting of the query q, candidate row subgraphs $\{\mathcal{G}^*(\rho_i)\}_{i=1}^{k}$, and retrieved passage texts $\mathcal{D}_{\text{L1}}$. Let a^* denote the target answer to query q.

Suppose Tier 1 evaluation produces an unresolved diagnostic check ψ_{check} and a focus candidate set $\mathcal{C}_{\text{focus}} \subset R$. By the definition of the targeted diagnostic verification in Equation (10), any discarded row $r \in R \setminus \mathcal{C}_{\text{focus}}$ is deterministically refuted by L1 (i.e., satisfies $\neg\psi_{\text{cand}}(r)$ under verified cell constraints), which implies:

$$P(a^* \in \text{Attr}(r) \mid q, \psi_{\text{check}}) = 0 \quad \forall r \notin \mathcal{C}_{\text{focus}}. \tag{18}$$

Hence, the support of the conditional distribution of a^* is strictly restricted to the attributes of the focus candidate set $\mathcal{C}_{\text{focus}}$ and their linked external passages:

$$a^* \in \bigcup_{c \in \mathcal{C}_{\text{focus}}} \left(\text{Attr}(c) \cup \text{Ent}(\mathcal{D}_c)\right). \tag{19}$$

Let $\mathcal{X}_{\text{discard}} = \mathcal{X}_{\text{L1}} \setminus (\{q, \psi_{\text{check}}\} \cup \{\mathcal{G}^*(\rho_c)\}_{c \in \mathcal{C}_{\text{focus}}})$ denote the discarded context comprising non-candidate rows and unlinked passages. Because a^* is conditionally independent of $\mathcal{X}_{\text{discard}}$ given the true candidate partition $\mathcal{C}_{\text{focus}}$ and the resolving supplemental text $\Delta\mathcal{D}$ that grounds ψ_{check}, we have:

$$P(a^* \mid \mathcal{X}_{\text{L1}} \cup \Delta\mathcal{D}, \mathcal{K}_{\text{L2}}) = P(a^* \mid \mathcal{K}_{\text{L2}}), \tag{20}$$

where $\mathcal{K}_{\text{L2}} = (q,\, \psi_{\text{check}},\, \{\mathcal{G}^*(\rho_c)\}_{c \in \mathcal{C}_{\text{focus}}},\, \Delta\mathcal{D})$ as defined in Equation (11).

By the definition of conditional mutual information:

$$\begin{aligned} I(a^*; \mathcal{X}_{\text{L1}} \cup \Delta\mathcal{D} \mid \mathcal{K}_{\text{L2}}) &= H(a^* \mid \mathcal{K}_{\text{L2}}) - H(a^* \mid \mathcal{K}_{\text{L2}}, \mathcal{X}_{\text{L1}} \cup \Delta\mathcal{D}) \\ &= H(a^* \mid \mathcal{K}_{\text{L2}}) - H(a^* \mid \mathcal{K}_{\text{L2}}) = 0. \end{aligned} \tag{21}$$

Therefore, the state capsule $\mathcal{K}_{\text{L2}}$ preserves all answer-bearing mutual information, forming a sufficient statistic for a^* while pruning irrelevant non-candidate evidence. □